

Enhancing Π_0 with Spatial Knowledge

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1 Abstract

We enhance Physical Intelligence’s Π_0 vision-language-action model with 3D spatial understanding by integrating geometric scene features from Meta AI’s VGGT through LoRA fine-tuning. Our approach extracts encoder-level representations from the VGGT model and incorporates them into Π_0 ’s architecture. Experiments on the **libero-spatial** suite demonstrate that our method achieves a 20.6% success rate, significantly outperforming the 4.6% baseline, despite evaluation constraints that limited access to full runtime geometric features.

2 Introduction

Vision-language-action (VLA) models have shown promise in robotic manipulation, but often struggle with tasks requiring precise spatial reasoning. While models like Π_0 [1] demonstrate strong performance on language-conditioned tasks, achieving success rates exceeding 90% on standard benchmarks like LIBERO[2], recent robustness evaluations reveal vulnerabilities in their spatial understanding. Specifically, the LIBERO-PRO benchmark[3] shows that Π_0 ’s performance can collapse to nearly 0% when object positions are perturbed, highlighting a reliance on 2D visual pattern matching rather than geometric reasoning. Similarly, findings from the LIBERO+ benchmark[4] indicate that Π_0 ’s success rates can drop by over 75% under camera viewpoint changes.

Recent work in SpatialVLA[5] and Evo-0[6] demonstrates that incorporating 3D scene representations can improve manipulation performance, with Evo-0 showing that multi-view camera observations provide sufficient information for reconstructing 3D scene geometry.

We build upon these insights by integrating VGGT[7], a state-of-the-art 3D reconstruction model, with Π_0 . Our approach extracts latent geometric features from VGGT’s encoder and incorporates them into Π_0 through LoRA fine-tuning, enabling the model to leverage rich spatial representations while maintaining computational efficiency. This work uses the **openpi**

repository¹ for its LoRA fine-tuning tools and LIBERO benchmark integration.

3 Related Work

Vision-Language-Action Models. Π_0 [1] represents a significant advance in VLA models, demonstrating strong zero-shot generalization across diverse manipulation tasks through pre-training on large-scale robot data and vision-language alignment.

3D Spatial Representations for Manipulation. SpatialVLA[5] demonstrates that incorporating 3D geometric features improves performance on spatially-grounded tasks. Evo-0 [6] shows that multi-view RGB observations from robot cameras contain sufficient information for 3D scene reconstruction, motivating camera-based approaches over depth sensors.

3D Scene Reconstruction. VGGT [7] achieves state-of-the-art performance in reconstructing 3D scenes from multi-view images through iterative aggregation of visual features. We leverage VGGT’s encoder to extract spatial representations for manipulation tasks.

4 Proposed Method

4.1 Data Pre-Processing

To maximize training efficiency, we precompute VGGT features offline before fine-tuning. Our preprocessing pipeline extracts key RGB observations from each trajectory in **libero-spatial** and processes them through VGGT’s encoder to obtain spatial representations. These features are then aligned temporally with the original dataset trajectories and stored as an augmented version of **libero-spatial**.

The modifications described in this section are implemented in the `vggt/precompute_3d_features.py` file.

¹<https://github.com/Physical-Intelligence/openpi>

4.2 Fine-Tuning

Training is conducted on Google Cloud Platform using an L4 GPU. Due to memory constraints, we use a maximum batch size of 2. We evaluate two approaches for integrating VGGT representations:

- **Π_0 -VGGT Features:** Our initial approach extracts the full decoded outputs from VGGT, including 3D scene points, camera pose estimates, and depth maps. While these provide explicit geometric information, this approach has two limitations: (1) it requires running VGGT’s complete forward pass, including all decoder heads, and (2) it converts rich learned representations into explicit geometric quantities, potentially discarding latent information useful for manipulation.
- **Π_0 -VGGT Latents:** To address these limitations, we extract features directly from VGGT’s encoder by using only the aggregator module and selecting the final iteration’s aggregated tokens. This encoder-only approach is computationally faster and preserves the rich latent scene understanding learned by VGGT.

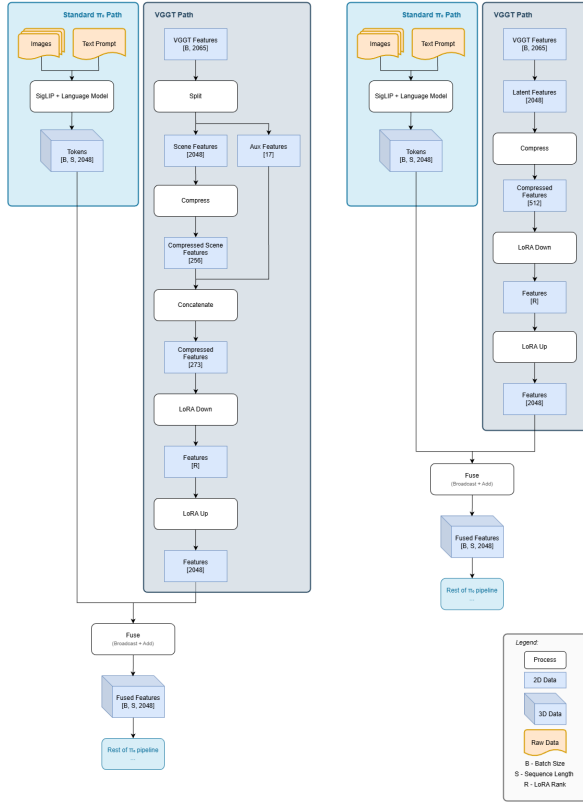


Figure 1: Π_0 -VGGT Features (left) vs Π_0 -VGGT Latents (right)

The implementation details for the fine-tuning process can be found in

`openpi/src/openpi/training/libero_vggt_loader.py`,
`openpi/src/openpi/models/pi0_with_vggt.py`,
`openpi/src/openpi/models/pi0_config.py`,
`openpi/src/openpi/training/config.py`.

5 Data

5.1 Fine-Tuning Dataset

We fine-tuned our model exclusively on the `libero-spatial` dataset. While this decision was influenced by time and resource constraints, this dataset was chosen because it directly evaluates the spatial reasoning skills our method targets.

5.2 Evaluation Benchmarks

Evaluation was conducted using the LIBERO benchmark[2], utilising the `libero-spatial` suite for in-distribution spatial reasoning and the `libero-10` suite to assess multitask generalisation.

To assess robustness and compare model performance under complex conditions, we utilised the LIBERO+ benchmark[4]. This suite extends the standard evaluation by introducing perturbations across seven dimensions, including variations in robot arm initial states, camera viewpoints, lighting, and object layouts. These modifications allow us to evaluate the extent to which model performance degrades when subjected to environmental shifts that are not present in the clean training data.

To obtain a more detailed understanding of whether the proposed solution enables improved spatial reasoning, we additionally evaluated both a baseline and a VGGT-enhanced model on a modified `libero-spatial` suite, where the agent camera’s position and viewing angle were altered during evaluation to test robustness under viewpoint changes. The detailed angles tested can be found in Table 1.

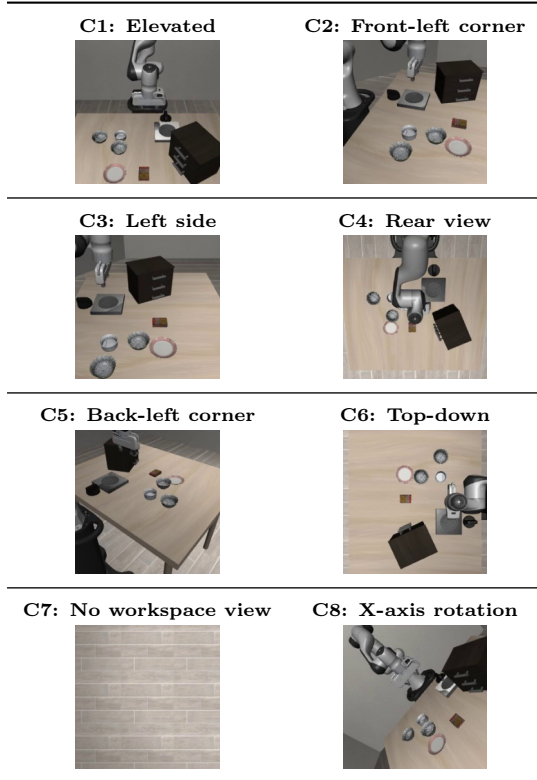
The evaluation described in this section is implemented in the `openpi/examples/libero/main.py` file.

6 Experimental Results

6.1 Metrics

To evaluate both the baseline and enhanced models, we selected two metrics. The first is the average success rate of the model across all tasks and episodes. The second metric counts the average number of movements required to accomplish a task (calculated only for successful episodes), where a lower number indicates better efficiency.

Table 1: Agent camera view used during evaluation with modified spatial suite.



6.2 Results

We report success rates across the LIBERO benchmark suites. Model configurations are abbreviated as follows: **B** (Batch Size), **R** (LoRA Rank), and **k** (Training Steps $\times 10^3$). Note that, for the LIBERO evaluation, 50 trials per task were used, while for LIBERO+, only 1 trial per task was used, as per its usage instructions². The latter evaluation was reduced to two models due to its lengthy runtime (~ 67 h).

Table 2 summarises the average success rates among all episodes per task, for each model.

Table 2: Success rate (%) on LIBERO benchmarks.

Model Configuration	Libero Spatial (%)	Libero 10 (%)	Libero+ Spatial (%)
Baseline (B2, 5k)	0.4	0.0	-
Baseline (B2, 30k)	4.6	-	2.04
Π_0 -VGGT features, R8 (B2, 5k)	2.0	0.0	-
Π_0 -VGGT latents, R8 (B2, 5k)	0.4	0.0	-
Π_0 -VGGT latents, R8 (B2, 30k)	8.8	0.0	0.0
Π_0 -VGGT latents, R16 (B2, 30k)	20.6	0.0	-

Figure 2 and Table 3 detail the successful episodes per task for each model.

Finally, Tables 4 and 5 summarise, respectively, general and task-specific metrics for the modified **libero-spatial** suite, using the notation described in Table 1.

²<https://github.com/sylvestf/LIBERO-plus>

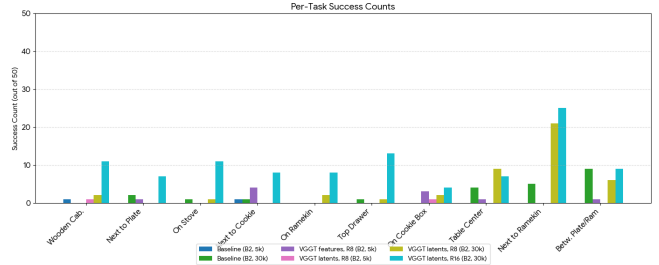


Figure 2: Per-task successful episodes (out of 50).

Table 3: Per-task successful episodes (out of 50).

Task	Baseline		Feat. R8 5k	Π_0 -VGGT Latents		
	5k	30k		R8 5k	R8 30k	R16 30k
Wooden Cab.	1	0	0	1	2	11
Next to Plate	0	2	1	0	0	7
On Stove	0	1	0	0	1	11
Next to Cookie	1	1	4	0	0	8
On Ramekin	0	0	0	0	2	8
Top Drawer	0	1	0	0	1	13
On Cookie Box	0	0	3	1	2	4
Table Center	0	4	1	0	9	7
Next to Ramekin	0	5	0	0	21	25
Betw. Plate/Ram	0	9	1	0	6	9

6.3 Discussion

Our results demonstrate that incorporating VGGT’s encoder-level latent representations substantially improves Π_0 ’s performance on spatially-grounded manipulation tasks. The Π_0 -VGGT Latents model with rank-16 LoRA achieved 20.6% success rate on **libero-spatial**, representing a $4.5\times$ improvement over the 30k-step baseline (4.6%).

At 5k steps, the Π_0 -VGGT Features model outperformed the Latents configuration (2.0% vs. 0.4%), indicating superior early-stage performance. However, due to the significant computational overhead of the Features approach, we prioritized the more efficient Latents method for extended training.

The substantial performance gap between rank-8 (8.8%) and rank-16 (20.6%) configurations suggests that spatial feature integration benefits from higher model capacity. Task-specific analysis reveals considerable variation, with the model excelling at precise localization tasks like "Next to Ramekin" (25/50 successes) while struggling with others like "On Cookie Box" (4/50).

A significant limitation emerges in robustness: the

Table 4: Number of successes and average number of movements before success on modified suite (20 trials for each task).

Metric	Model	C1	C2	C3	C4	C5	C6	C7	C8	Default
Successes	Baseline (B2, 30k)	12	11	11	15	13	6	9	10	9
	Π_0 -VGGT latents, R16 (B2, 30k)	40	43	43	56	45	43	54	51	37
Avg. Steps	Baseline (B2, 30k)	172.2	160.9	174.7	161.6	168.6	145.0	173.0	174.7	150.3
	Π_0 -VGGT latents, R16 (B2, 30k)	154.0	153.4	159.7	147.2	146.4	157.9	157.0	152.8	151.9

Table 5: Task-specific success counts on modified suite (20 trials for each task).

Model	Task Description	C1	C2	C3	C4	C5	C6	C7	C8	Default
Baseline (B2, 30k)	Wooden Cab.	0	0	0	0	0	0	0	0	0
	Next to Plate	0	0	0	2	3	1	0	1	2
	On Stove	0	0	0	2	0	0	0	0	0
	Next to Cookie	0	0	0	0	0	0	0	0	0
	On Ramekin	0	0	0	0	0	0	0	0	0
	Top Drawer	0	2	0	1	0	0	0	0	0
	On Cookie Box	1	1	0	3	1	0	0	1	0
	Table Center	2	2	2	5	1	2	1	4	1
	Next to Ramekin	3	2	5	2	4	1	5	2	4
	Betw. Plate/Ram	6	4	4	0	4	2	3	2	2
Π_0 -VGGT latents, R16 (B2, 30k)	Wooden Cab.	7	7	3	5	2	4	6	5	5
	Next to Plate	4	3	3	7	6	1	3	4	2
	On Stove	3	7	3	9	4	6	7	8	8
	Next to Cookie	1	1	3	3	2	3	5	2	2
	On Ramekin	5	5	6	5	6	6	6	1	3
	Top Drawer	4	4	8	7	5	3	7	7	6
	On Cookie Box	0	3	0	1	0	2	2	2	0
	Table Center	7	4	7	7	7	7	6	5	3
	Next to Ramekin	7	8	8	10	9	8	8	13	5
	Betw. Plate/Ram	2	1	2	2	4	3	4	4	3

0% success rate on LIBERO+ for the R8 30k model, compared to 2.04% for the baseline, indicates that while VGGT features improve nominal performance, they may encode viewpoint-specific or lighting-dependent information that harms generalization under distribution shift. This suggests that current VGGT features require additional techniques, such as multi-view fusion or data augmentation, to achieve robust performance across environmental variations.

The evaluation on **libero-10** provides no additional conclusions other than confirming that our training conditions were too constrained to achieve good model generalization, which is another factor to consider regarding our previous findings.

The evaluation on the modified **libero-spatial** suite confirms that the Latent-aware model significantly outperforms the baseline, increasing success rates across all viewpoints. Additionally, it demonstrates superior efficiency, achieving success in fewer steps on average. Notably, the model integrating VGGT performs surprisingly well even when no information can be extracted from the agent camera at startup. This suggests it may, in this case, prioritize the wrist view or rely on a layout learned during training.

6.4 Limitations

Due to time and resource constraints, our evaluation setup introduced several train-test mismatches. The Π_0 -VGGT Features **libero-spatial** evaluation was conducted without runtime VGGT feature computation, relying instead on standard RGB observations. The Π_0 -VGGT Latents evaluation (Tables 2, 3) similarly lacked runtime VGGT latents integration, while the modified spatial suite evaluation (Tables 4, 5) incorrectly used the Features model inputs instead of Latents. These inconsistencies mean our reported performance represents a lower bound on the true capabilities of VGGT-augmented mod-

els. Remarkably, the Latents model still achieved 20.6% success despite feature ablation, suggesting that VGGT-augmented training improved spatial reasoning capabilities even when geometric features are unavailable at test time.

Finally, data regarding the number of movements before success is missing for the standard suites. We failed to report this metric due to an issue in our result-tracking pipeline. We acknowledge this as an avoidable oversight; however, it does not impact the study’s overall qualitative conclusions.

7 Conclusion and Future Work

We presented an approach to enhance vision-language-action models with 3D spatial understanding by integrating VGGT’s encoder-level latent features into Π_0 through LoRA fine-tuning. Our results demonstrate that preserving VGGT’s rich learned representations yields a $4.5\times$ improvement over baseline on **libero-spatial**, achieving 20.6% success rate.

However, degraded LIBERO+ performance reveals important robustness limitations. Future work should explore data augmentation during feature extraction, multi-view VGGT fusion for viewpoint-invariant representations, and alternative fusion mechanisms, such as cross-attention. Scaling experiments with larger batch sizes, higher LoRA ranks, and evaluation on additional LIBERO suites would provide further insights. Validating the approach on physical robot systems remains critical for assessing practical utility, where sim-to-real transfer and sensor noise present additional challenges beyond our simulation-based evaluation.

References

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